CIE

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FORM

TITLE: Lintellect- Code Review Agent

SCHOOL: Chitkara University, Rajpura, Punjab

DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO SHARE LOGIN AND PASSWORD WITH MENTORS

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ABSTRACT

1.1 Problem Background

Modern software development cycles demand rapid iteration and high-quality code, yet manual code review processes remain time-consuming and inconsistent. Teams often struggle with delayed reviews, overlooked edge cases, and inconsistent adherence to coding standards. Junior developers lack immediate feedback, leading to repeated mistakes and technical debt accumulation. Existing static analyzers detect syntax issues but fail to provide contextual reasoning or architectural insights. Furthermore, review feedback is often subjective and varies across reviewers. These limitations create bottlenecks in deployment pipelines, reduce productivity, and increase the risk of bugs reaching production environments.

1.2 Objective of the Project

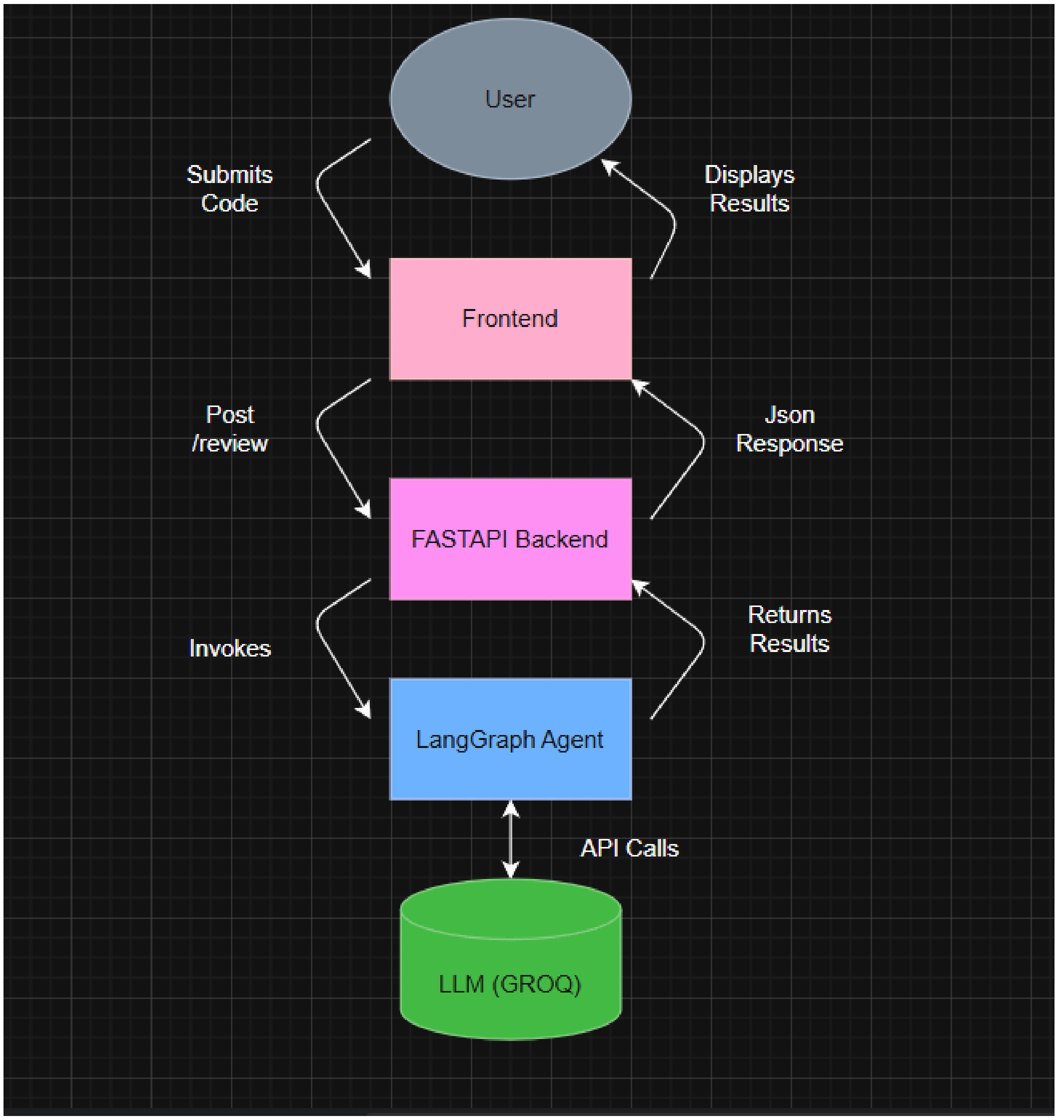
The objective of this project is to develop **Lintellect**, an AI-powered code reviewer that delivers instant, contextual, and intelligent feedback on source code by detecting logical errors, security vulnerabilities, performance inefficiencies, and best-practice violations while improving overall code quality and developer productivity.

1.3 Proposed Solution

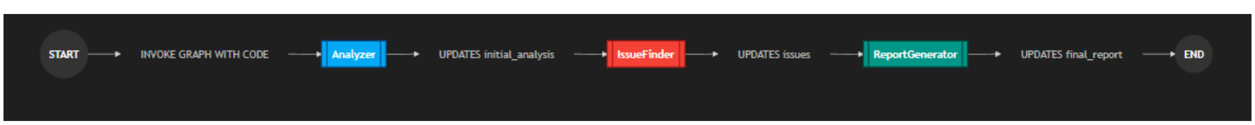
Lintellect is an intelligent code analysis platform that integrates with development workflows through repository access or direct code input interfaces. The system processes source code using advanced AI-driven analysis models combined with rule-based static analysis techniques. It evaluates syntax, structure, logic flow, security patterns, and performance concerns, generating actionable feedback with explanations and improvement suggestions. The major components include a Code Parsing Engine for language-aware analysis, an AI Review Engine for contextual evaluation, a Rule-Based Validation Module for coding standards enforcement, and a React-based Interactive Dashboard that displays review summaries, categorized issues, and improvement recommendations. These components collectively provide automated, scalable, and consistent code review capabilities.

1.4 System Design and Workflow

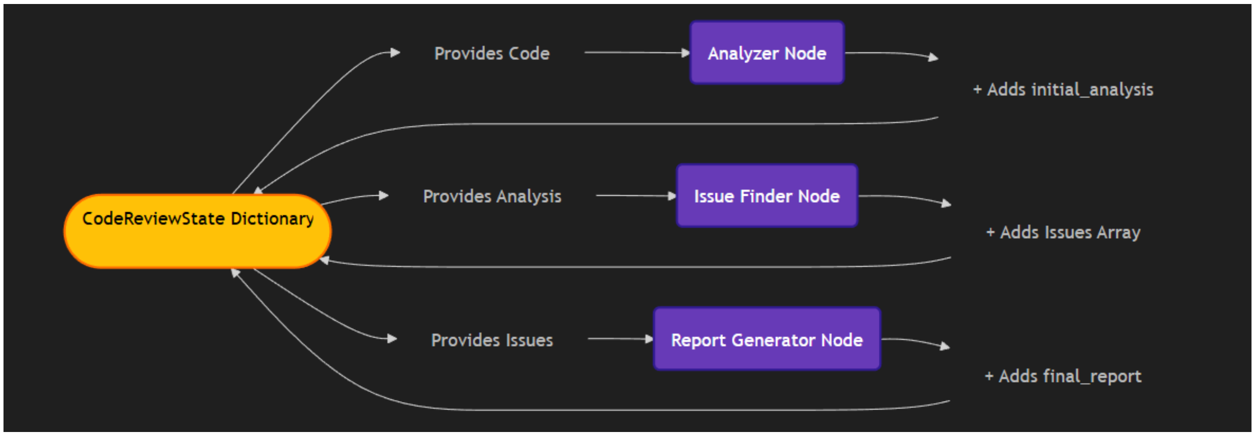
1.4.1 High Level System Architecture



1.4.2 LangrGraph Workflow



1.4.3 State Data Flow



1.5 Key Features or Functionality

Key features include intelligent bug detection, performance optimization suggestions, security vulnerability identification, and best-practice enforcement. Lintellect provides categorized feedback such as critical errors, warnings, and improvement suggestions. It supports multiple programming languages with real-time review capabilities. Inline code explanations help developers understand issues clearly. The platform generates structured review reports for documentation and collaboration. Integration-ready APIs allow incorporation into CI/CD pipelines for automated pre-deployment checks.

1.6 Technology / Method Used

The system utilizes Python 3.x with FastAPI for backend APIs, AI-based analysis models for contextual code review, React for interactive frontend dashboards, Tailwind for design.

1.7 Expected Outcome & Benefits

The expected outcome is a production-ready AI reviewer that reduces manual review time by over 70%, improves early bug detection, and standardizes coding practices across teams. Lintellect enhances developer learning through instant feedback, reduces technical debt accumulation, and accelerates deployment cycles. Organizations benefit from improved code reliability, stronger security posture, and increased development efficiency without increasing review workload.

1.8 Original Contribution Statement

This project presents an original contribution through a contextual AI-driven code evaluation engine, multi-layered validation architecture, structured feedback categorization, and automated review reporting optimized for scalable integration into modern development workflows.